CX-01 — What a modern "AI portal chat" workspace looks like

Research for the Spiff mockup. Focus: Claude Cowork + the wider vocabulary of agentic chat workspaces. Everything below is sourced; where a claim could not be verified from a primary source it is flagged [unverified].

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1. Claude Cowork — what it is

Cowork is Anthropic's agentic mode for non-developers — a system that "execute[s] multi-step knowledge work on a user's behalf," aimed at "researchers, analysts, operations teams, legal professionals, finance teams." The framing is explicitly not conversational: "It moves between them, synthesizes information across multiple sources, and completes tasks without the user coordinating each step."
([anthropic.com/product/claude-cowork](https://www.anthropic.com/product/claude-cowork))

- Chat and Cowork are one interface, not two apps. As of July 2026 the mode selector lives in the message box: "In the message box, select 'Cowork'," and select "Chat" to go back.
([support — Get started](https://support.claude.com/en/articles/13345190-get-started-with-claude-cowork); [7 Jul 2026 merge](https://cryptobriefing.com/anthropic-claude-chat-cowork-interface-update/)). Earlier builds used Chat / Cowork / Code tabs at the top of Claude Desktop ([DataCamp](https://www.datacamp.com/tutorial/claude-cowork-tutorial)) — the same product has oscillated between "tab" and "mode-in-composer."
- Surfaces: desktop, web, mobile, Chrome side panel (which "starts a Cowork session directly, with no selector"). Live artifacts are desktop-only. Sessions are portable: "Resume a session started on another surface."
([support — web, desktop, mobile](https://support.claude.com/en/articles/15520349-use-claude-cowork-on-web-desktop-and-mobile))
- GA for paid plans 10 April 2026, with "role-based access controls for teams, spend limits and use controls for governance, analytics and visibility, and integrations and plugins."
([TechRadar](https://www.techradar.com/pro/claude-cowork-is-now-available-for-enterprise-use-adds-analytics-access-controls-and-more))

1.1 The three-panel task anatomy

The single most useful UI finding. A Cowork task shows three stacked panels on the right:

Panel	Answers	Contents
Progress	"what Claude will do"	Numbered plan; done steps get a checkmark + strikethrough, current step highlighted, future steps plain
Project	"what can Claude see?"	Instructions · CLAUDE.md, project files, and a Scratchpad audit trail marking each file "wrote to", "viewed", "created"
Context	"what can Claude reach?"	Uploads (this task only) and Connectors (active integrations only — installed-but-inactive ones don't appear)

([Camp Claude — Anatomy of a Cowork Task](https://camp-claude.github.io/learn/cowork-task-anatomy/))

Corroborated: a "visible todo list" you "review before it starts. Each step gets checked off as it completes" ([aimaker](https://aimaker.substack.com/p/claude-cowork-review-agentic-ai-guide)); and a
"Progress sidebar on the right displaying real-time task updates," an "Artifacts pane showing files

Claude reads or creates, clickable for preview," plus a terminal output view ([DataCamp](https://www.datacamp.com/tutorial/claude-cowork-tutorial)).

The todo lifecycle is documented: pending → in_progress → completed (→ deleted), with a separate activeForm label ("Identifying newsletter threads") shown instead of the static subject while a step runs. Todos are created for "complex multi-step tasks requiring three or more distinct actions." ([Track todos](https://code.claude.com/docs/en/agent-sdk/todo-tracking)) This is the canonical spec for an agent progress list — copy its state machine.

1.2 Approval modes (the governance control)

A mode selector in the chat box: Manual ("Manually approve") pauses for approval; Auto ("Automatically approve") works autonomously but "reviews each action for safety"; Skip ("Skip all approvals"). Users can steer mid-run — "jump in to course-correct." ([support — Get started](https://support.claude.com/en/articles/13345190-get-started-with-claude-cowork))

Admins gate these at Organization settings > Cowork: "Enable for your organization," "Run Cowork in the cloud," "Allow 'Automatically approve' mode," "Allow 'Always allow' for connector tools" (off by default). Enterprise adds groups + custom roles; events stream via OpenTelemetry to SIEM covering "tool calls, file access, human approval decisions"; remote sessions are retrievable via the Compliance API. ([support — Team and Enterprise](https://support.claude.com/en/articles/13455879-use-claude-cowork-on-team-and-enterprise-plans))

1.3 Projects, folders, memory

Projects sit in the left nav, created with the "+" next to "Projects," each with "their own files, context, instructions, and memory." Components: Instructions ("Add tone, formatting, or rules"), Context (folders, chat projects, URLs), Scheduled tasks, Memory — "Memory is scoped to the project, so what Claude learns in one project doesn't carry over to others." Created from scratch, imported from a chat project, or from an existing folder. Cowork projects are desktop-only and stored locally. ([support — Projects in Cowork](https://support.claude.com/en/articles/14116274-organize-your-tasks-with-projects-in-claude-cowork))

Folder access is explicit: a "Work in a Folder" checkbox plus a dialog asking to "read, edit, and delete files" with one-time or "Always Allow" ([DataCamp](https://www.datacamp.com/tutorial/claude-cowork-tutorial)). Standing rules live at Settings > Cowork → Global instructions → Edit.

1.4 Scheduled tasks

"Scheduled" in the left sidebar → "New task" → "Create with Claude" (clarifying questions with multiple-choice answers, then a proposed name + schedule + description you confirm with "Schedule") or "Set up manually" (task name, prompt, approval mode, frequency — hourly / daily / weekly / weekdays / manually — optional model, optional folder). The page shows all tasks plus upcoming and past runs, with edit, pause, resume, delete, run-on-demand. /schedule works inline. Scheduled tasks "run remotely, so they run on their cadence even when your computer is asleep." ([support — Schedule recurring tasks](https://support.claude.com/en/articles/13854387-schedule-recurring-tasks-in-claude-cowork))

1.5 Live artifacts

"Persistent, interactive HTML dashboards" that refresh with current data from connected apps and keep version history — each "marked with a 'Cowork' label." They live in the Artifacts view with a "Live artifacts" tab, a "Filter by" dropdown top-right, a refresh button in the header, and version history. Created from a task or via "New artifact" → "Create Cowork artifact."

The governance detail worth stealing outright: sharing is Team/Enterprise only, via "Share" → "Share & copy link", "Sharing stays within your organization" with "no external or public links," and "Viewers use their own connector access, not the creator's."
([support — Live artifacts](https://support.claude.com/en/articles/14729249-use-live-artifacts-in-claude-cowork))

1.6 Dispatch, notifications, the built-in browser

Dispatch messages Claude from phone or desktop; "Claude figures out what kind of work is needed and spins up the right session… These sessions appear in their respective sidebars." It deliberately hides the trace: "Claude messages you the outcome (a spreadsheet, a memo, a comparison table, a pull request) rather than showing you every step." Dispatch is one continuous thread with no thread management; push fires "when a task is done or when Claude needs your go-ahead."
([support — Assign tasks from anywhere](https://support.claude.com/en/articles/13947068-assign-tasks-from-anywhere-in-claude-cowork))

Built-in browser: "a browser opens in the side panel and Claude navigates webpages, reads them, clicks, and types," while you keep working. "Claude never sees your tabs, bookmarks, or passwords"; per-site login import; banking/email/SSO excluded by default. Settings → Cowork → Preferred browser.
(claude.com/blog/cowork-built-in-browser)

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2. Claude's chat UI vocabulary

- Composer: the "+" button lower-left, or type "/" for commands. Model name shown below the input on web/desktop (top of screen on mobile); click it for the model selector and "model, effort, and thinking settings."
([support — Get started with Claude](https://support.claude.com/en/articles/8114491-get-started-with-claude))
- Thinking disclosure: a "Thinking" indicator with a timer, in an expandable section above the response, containing "Claude's thought process summary." Toggled from the "Search and tools" button lower-left.
([support — model, effort, thinking](https://support.claude.com/en/articles/10574485-using-extended-thinking))
- Research mode: "+" → "Research"; "A blue indicator will appear on the bottom of the chat window"; output has "easy-to-check citations."
([support — Use research](https://support.claude.com/en/articles/11088861-use-research-on-claude))
The step-by-step progress rendering of Research is [unverified] — support docs don't describe it.
- Artifacts: "a dedicated window to the right of the main chat," triggered by content that "is significant and self-contained, typically over 15 lines" and likely to be edited or reused. Has a version selector, "Edit with Claude" on highlighted markdown, lower-right controls to view code / copy / download, a slider icon upper-right for multiple artifacts in one conversation, and "Try fixing with Claude" on errors.
([support — Artifacts](https://support.claude.com/en/articles/9487310-what-are-artifacts-and-how-do-i-use-them))
- Sharing: Share button upper right → "Share"; produces a snapshot of messages sent

before sharing. Attachments and "raw data retrieved from MCP tool calls" stay private. Unshare via the visibility dropdown ("Public" → "Private"); audit at Settings > Privacy → Shared chats. Team/Enterprise can only share inside the org.
([support — Share and unshare chats](https://support.claude.com/en/articles/10593882-share-and-unshare-chats))
- Memory & search: retrieving a past chat "you will see this reflected in your current chat as a tool call." Settings > Memory has "Search and reference chats," "Generate memory from chats," and "Include sensitive topics in memory" — the last surfaces "a notice … above the message box" each time something sensitive is saved. Incognito = a ghost icon upper-right.
([support — chat search and memory](https://support.claude.com/en/articles/11817273-use-claude-s-chat-search-and-memory-to-build-on-previous-context))
- Keyboard/quick entry (macOS): double-tap Option to open Claude from any app; Option + Space alternative; Caps Lock for dictation; customised at Settings > General → "Desktop app."
([support — Quick entry](https://support.claude.com/en/articles/12626668-use-quick-entry-with-claude-desktop-on-mac))
A published full keyboard-shortcut map for claude.ai chat is [unverified].
- Usage indicators: support docs cover limits policy but do not document the in-product indicator UI — [unverified].
- Message actions (copy / retry / edit-and-branch): widely present in the product but not documented in the Conversation management collection, which covers only delete/rename, share, incognito, search/memory, model-switch notices — [unverified] as to exact labels.

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3. Skills, connectors, plugins as end-user surfaces

Skills live at Customize > Skills, split into Personal / Shared / Organization.
Prerequisite: enable "Code execution and file creation" (Settings > Capabilities, or Organization settings > Skills). Each skill is a toggle; "+" creates or uploads one; grayed-out = disabled or blocked by org settings; a "Share" button distributes it. Claude invokes them automatically; on some surfaces users type "/" to browse.
([support — Use skills](https://support.claude.com/en/articles/12512180-use-skills-in-claude))

Connectors live behind the "+" button lower-left or "/" → hover "Connectors" → toggle per conversation. Org owners set Tool permissions per category (read-only vs write/delete) to "Always allow," "Needs approval," or "Blocked." A "Tool access" setting offers "Auto" (default) or "On demand." Connectors with an "Interactive" badge "render live interfaces — like dashboards, task boards, and design tools — directly within your conversation," as inline cards or fullscreen view, in sandboxed iframes.
([Connectors](https://support.claude.com/en/articles/11176164-use-connectors-to-extend-claude-s-capabilities);
[Interactive connectors](https://support.claude.com/en/articles/13454812-use-interactive-connectors-in-claude))

Plugins bundle "skills, connectors, and sub-agents into a single package": Cowork tab → Customize → Plugins → "Browse plugins" → "Install." Marketplaces (default "Knowledge Work") added via "+" → "Add marketplace." Note: "Hooks and sub-agents run only in Cowork, so they appear grayed out in chat."
([Use plugins](https://support.claude.com/en/articles/13837440-use-plugins-in-claude)). The financial-services marketplace is the worked domain example — a core "Financial analysis" plugin plus add-ons exposing /comps [company], /dcf [company], /ic-memo [project name].
([Claude Academy](https://academy.claude.com/tutorials/install-financial-services-plugins-for-cowork))

Enterprise search ships as a pre-configured project named "Ask Your Org," "starred in your sidebar by default," and is "Permission-aware: You only see search results from data you have permission to access in the original systems," with "Each user authenticates with their own

credentials."
([Enterprise search](https://support.claude.com/en/articles/12489464-use-enterprise-search))
The closest shipped analogue to Spiff's governance invariants.

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4. Comparable workspaces — distinctive UI moves

Product	Distinctive move	Source
ChatGPT canvas	Opens "in a separate window"; directly editable by the user; a contextual shortcuts menu (adjust length, reading level, add polish / review code, add logs, fix bugs); versions via a back button; inline suggestions rather than chat critique; opens automatically when detected useful	openai.com/index/introducing-canvas
ChatGPT agent	/agent in the composer or the tools menu; "'…' → Take over browser" hands control back to the human for passwords; Clock icon turns a finished run into a repeating schedule; central chatgpt.com/schedules page	[help.openai.com — ChatGPT agent](https://help.openai.com/en/articles/11752874-chatgpt-agent)
M365 Copilot (May 2026 redesign)	Expand/contract left nav over "agents, conversations, and history"; the prompt line becomes an expandable workspace with inline formatting; layered responses (plain answer first, structure and next-step actions added as you refine); in-app side pane as "an editing partner … with clear signals so you always know what it's doing"; capability agents (Designer, Researcher, Word, Excel, PowerPoint)	[microsoft.com blog](https://www.microsoft.com/en-us/microsoft-365/blog/2026/05/28/introducing-a-new-design-for-microsoft-365-copilot/); [support.microsoft.com — agents](https://support.microsoft.com/en-us/topic/using-agents-for-microsoft-365-copilot-169469d7-328d-4d37-9090-bfc2058a39bd)
Gemini side panel (Docs)	"Ask Gemini" top-right; auto-summarises the open doc on open; suggested prompts + "More suggestions"; @ to reference Drive files, or "Sources" → "Add from Drive"; per-response Insert / Retry / Preview / Copy buttons; thumbs feedback	[support.google.com](https://support.google.com/docs/answer/14206696?hl=en)
Notion Agent	Persistent "friendly face at the bottom"; "Switch chat mode" toggles sidebar vs floating window; results render as an interactive table in chat; pin a chat to the top of the Chat tab; clock icon for chat history; model dropdown in-panel; a private "My Notion AI" page holds instructions + memory	[notion.com/help/notion-agent](https://www.notion.com/help/notion-agent)
Notion Custom Agents	Built by describing them in plain language; schedule or event triggers; "Every run is logged, so changes are visible and reversible"	[notion.com/releases/2026-02-24](https://www.notion.com/releases/2026-02-24)
Perplexity Projects (formerly Spaces)	"A persistent, shareable workspace" holding conversations, computer tasks, files, custom instructions, connected tools; a Files section; a Settings tab for instructions and prioritised domains; a Brain tab for accumulated memory; roles Owner / Can edit / Can view and visibility Restricted / Organization can view / Anyone with the link	[perplexity.ai help](https://www.perplexity.ai/help-center/en/articles/10352961-what-are-spaces)

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5. Anatomy of a portal chat workspace

An exhaustive component inventory, assembled from the sources above.

A. Left rail (persistent nav)
1. Product/org logo + workspace switcher.
2. New task / New chat primary button.
3. Global search (Claude: ask naturally, rendered as a tool call).
4. Mode entries — in Claude these are now composer modes, but the rail still holds destinations: Chats, Cowork/Tasks, Code.
5. Projects section with a "+" to create.
6. Scheduled (tasks page).

7. Artifacts view (with a Live artifacts tab and Filter by).
8. Customize → Plugins / Skills / Connectors, consolidated "in one place."
9. Starred/pinned items (e.g. "Ask Your Org" starred by default).
10. Account, plan, settings, usage.

B. Thread list with grouping
Recents grouped by time; grouping by project; pin-to-top; rename/delete; incognito (ghost icon); search across history; per-thread status when a run is in flight; last-run timestamp for scheduled items.

C. Composer
Multiline expandable input (M365 makes it "an expandable workspace" that keeps pasted structure);
mode selector (Chat / Cowork) bottom-left; approval-mode selector (Manual / Auto / Skip); "+" menu for attachments, connectors, tools, Research; "/" command palette for skills and plugin commands (/schedule, /dcf); folder scope control ("Work in a Folder"); model picker with effort/thinking settings shown under the input; attachment chips; send/stop.

D. Message stream
User turns (editable); assistant turns; collapsible "Thinking" block with elapsed timer; tool calls rendered as inline chips/rows; permission-request cards with Allow once / Always Allow / Deny; interactive inputs (multiple choice, multi-select, ranking) so the user can click rather than type; inline cards from interactive connectors; citations; visual/interactive content built inline; message actions (copy, retry, feedback); consent notices above the message box.

E. Tool-trace layer
Per-step chips naming the tool and target; the todo state machine (pending / in_progress with an activeForm verb phrase / completed / deleted); a running "N of M completed" summary; a Scratchpad-style file audit ("wrote to", "viewed", "created"); terminal/browser output views; a side-panel browser you can watch.

F. Artifact / canvas panel
Right-hand dedicated window; tabs when several artifacts exist (slider icon); version selector and restore; direct user editing; "Edit with Claude" on a selection; shortcuts menu of common transforms; refresh button for live data; view-code / copy / download; Share & copy link; error-recovery button.

G. Task / progress panel
The numbered plan reviewable before execution; checkmarks and strikethrough; highlighted current
step; steer/interrupt; pause/resume; elapsed time; completion notification.

H. Right context panel
Project (instructions file, files, scratchpad) and Context (uploads scoped to this task, active connectors only). Plus memory/Brain, and per-project instructions.

I. Share / publish
Snapshot-based share links; org-only vs public visibility dropdown; a Shared audit list in settings; roles (Owner / Can edit / Can view); viewers use their own access, not the creator's.

J. Empty states
Task suggestions on entering the mode ("Organize files", "Crunch data"); Gemini's suggested prompts
+ "More suggestions"; M365's prompt starters per agent; guided setup flows for connectors.

K. Keyboard & OS surfaces
Global quick entry (double-tap Option), Option+Space, Caps Lock dictation, "/" palette, mobile push, browser side panel.

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6. Applying it to a reporting agent

Component	Reporting-agent mapping
Left rail Cowork/Tasks	"Answers" — every question a user asks becomes a durable, revisitable run, not an ephemeral chat
Projects	Reporting areas (e.g. GST Compliance, Revenue Assurance), each with its own instructions, allowed data domains, and memory. Memory scoped per project = no cross-domain leakage
Instructions · CLAUDE.md in the Project panel	The visible, editable "definitions file" — what revenue, active client, lodgement mean here. Governance made legible rather than hidden in a prompt
Context panel → Connectors (active only)	The data-source manifest for this answer: which warehouse, which schema, which reporting date. Show only what's actually reachable for this person
Approval-mode selector	Read-only questions run in Auto; anything that writes, emails, or publishes drops to Manual. The selector is the visible expression of "runs as its owner"
Progress panel	The query plan as a numbered, reviewable list: interpret question → resolve entities → check permissions → build query → validate against control totals → render. Reviewable before it runs is the trust-builder
Tool-trace chips	Each becomes an auditable lineage row: table touched, filter applied, rows returned, row-level security applied
Scratchpad audit ("viewed"/"wrote to")	The data-access log a compliance officer would ask for, shown to the user by default
Artifact / live artifact	The report itself — a live HTML dashboard with a refresh button and version history, so "the number changed" is explainable
Live-artifact sharing rule	The literal precedent for Spiff's core invariant: share the artifact, viewers use their own connector access, not the creator's. Frame the share dialog as "Recipients see this re-run against their own permissions"
Scheduled tasks	Recurring reports: "Set up manually" fields map almost 1:1 to a report schedule (name, prompt, approval mode, frequency, model, folder) — and the upcoming/past runs list is the run history
Sharing an automation	Per the invariant, the share action clones the schedule to the recipient, who becomes its owner — surface this as "Copy to my schedules," never "Add me as a viewer"
Skills / plugins	Report templates and house methodologies as / commands (/variance, /aging, /gst-return) — discoverable, org-published, toggleable, with grayed-out entries showing what governance has withheld
Enterprise search "Ask Your Org"	Precedent for a starred, pre-configured, permission-aware entry point — Spiff's front door
Interactive connectors (inline cards)	Filterable result tables inline in the answer, so a follow-up is a click not a re-prompt
Citations	Every figure carries a click-through to the query, the source table, and the as-at timestamp
Empty state	Question starters drawn from the user's actual entitlements — never suggest a report they cannot run
Usage/limits	Reframe as cost/scan transparency: rows scanned, warehouse credits, freshness of the underlying data

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7. Design lessons

1. Show the plan before the work. Cowork's highest-trust move is a numbered todo list the user reads and steers before execution. For Spiff that is the query plan — it turns black-box SQL into a reviewable proposal.
2. Use the documented state machine. pending → in_progress (verb-phrase activeForm) → completed. Present tense running, past tense done. Don't invent vocabulary.
3. Three panels answer three questions. Will do (Progress), can see (Project), can reach (Context). Separating capability from activity is what makes governance visible rather than

claimed.
4. List only what is actually live. Cowork's Context panel shows active connectors, not installed ones. Show the data this viewer can reach right now — an empty slot is itself an honest governance signal.
5. Make the audit trail a panel, not a log file. The Scratchpad's "viewed / wrote to / created" annotations are the pattern.
6. Put the risk dial in the composer. Approval mode sits beside the send button, per task, not buried in settings — and admins can remove the riskiest option entirely.
7. Governance is two layers. Per-conversation user toggles plus org-level permissions ("Always allow / Needs approval / Blocked", split read vs write). Mock both.
8. Re-scope on share, don't widen. "Viewers use their own connector access, not the creator's" is shipped, not aspirational. Put it in the share dialog's copy.
9. Share snapshots with holes by design. Claude's shared chats deliberately exclude attachments and raw MCP tool results. Share the narrative; re-run the numbers.
10. Collapse the trace by default, keep the timer visible. Presence of reasoning always shown, volume opt-in.
11. Deliverables outlive threads. An Artifacts view with tabs and a Filter-by dropdown treats outputs as a library. People return to the report, not the conversation.
12. Version history + refresh make a live number safe. Restore answers "what did we tell the board last month?"; refresh answers "is this current?"
13. Let the user click, not type. Multiple-choice / multi-select / ranking inputs and suggested-prompt chips reduce prompt anxiety — vital for non-analyst self-service.
14. Scheduling is the graduation path. Every good ad-hoc answer should offer "run this weekly" in one gesture (Clock icon / /schedule), then get a run-history page of upcoming and past runs.
15. Direct manipulation beats re-prompting. Editable canvases, "Edit with Claude" on a selection, and interactive inline tables all say: let the user grab the output.
16. Notify on two events only — done, or blocked-needing-approval. Deliver the outcome, not the trace.

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8. Confidence notes

- The three-panel Cowork anatomy comes from one detailed third-party source (Camp Claude), corroborated in outline by DataCamp and aimaker. Treat the exact panel headings as strong but not first-party.
- Message-action labels (retry / edit / branch), claude.ai keyboard shortcuts, usage-indicator UI, Research's in-progress rendering, and ChatGPT agent's live activity view are not documented in vendor help content — marked [unverified] above.
- Perplexity has renamed Spaces to Projects; the help centre now describes Projects.